

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 01 — INTRODUCTION TO PROGRAMMABLE LOGIC CONTROLLERS

Student Edition

Authorized ICDFA training environments only
icdfa.edu.ng

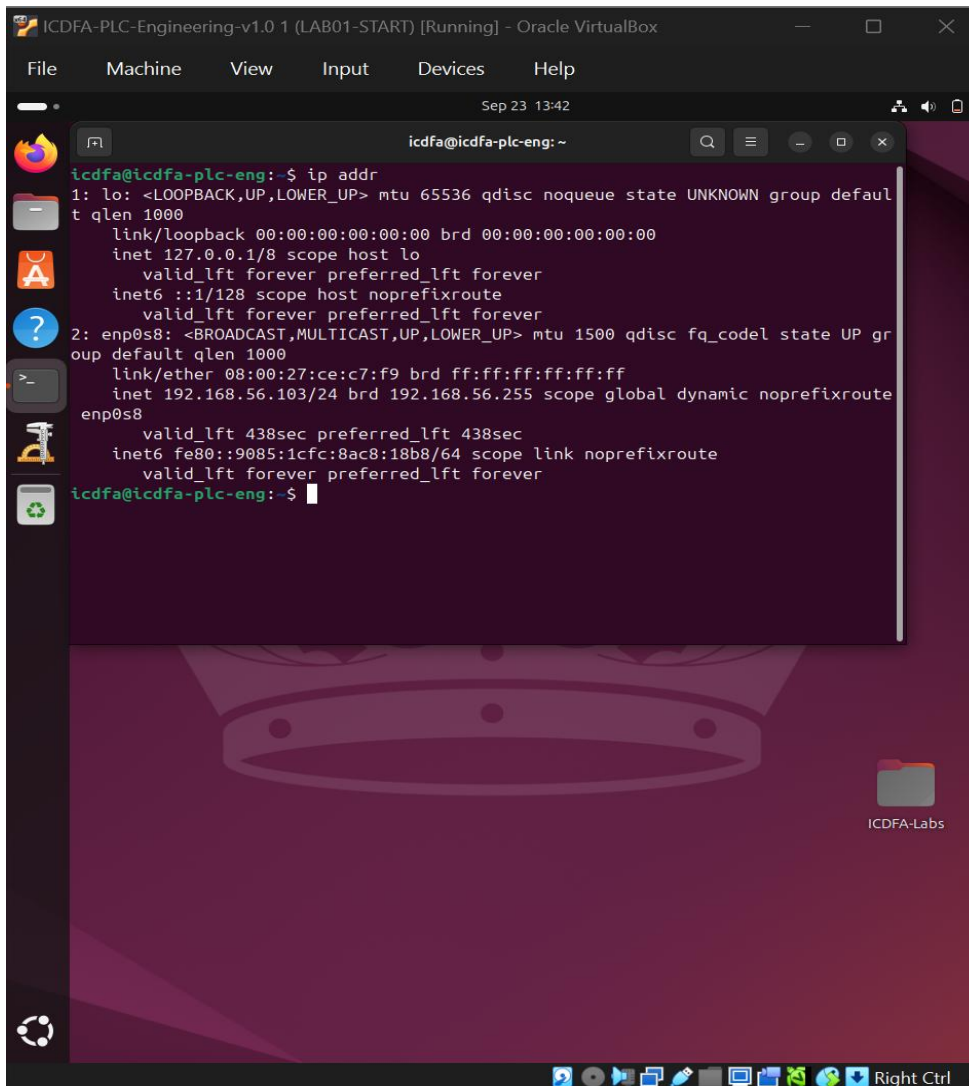
Lab 01: Introduction to Programmable Logic Controllers

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 2–3 hours

Submission deadline: As published in ICDFA E-Campus.

Required Evidence & Submission

- Screenshot showing the OpenPLC Motor_Control ladder diagram.
- A labelled explanation of the Start, Stop, latch and MotorOutput elements.
- A completed I/O classification table with at least four industrial examples.
- A 150–250 word reflection on why network-connected PLCs create both operational benefits and cybersecurity considerations.



The screenshot shows a terminal window titled "icdfa@icdfa-plc-eng: ~" with the following output for the command `ip addr`:

```
icdfa@icdfa-plc-eng:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ce:c7:f9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.103/24 brd 192.168.56.255 scope global dynamic noprefixroute
        valid_lft 438sec preferred_lft 438sec
    inet6 fe80::9085:1cfc:8ac8:18b8/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
icdfa@icdfa-plc-eng:~$
```

PLC VM Address: 192.168.56.103/24

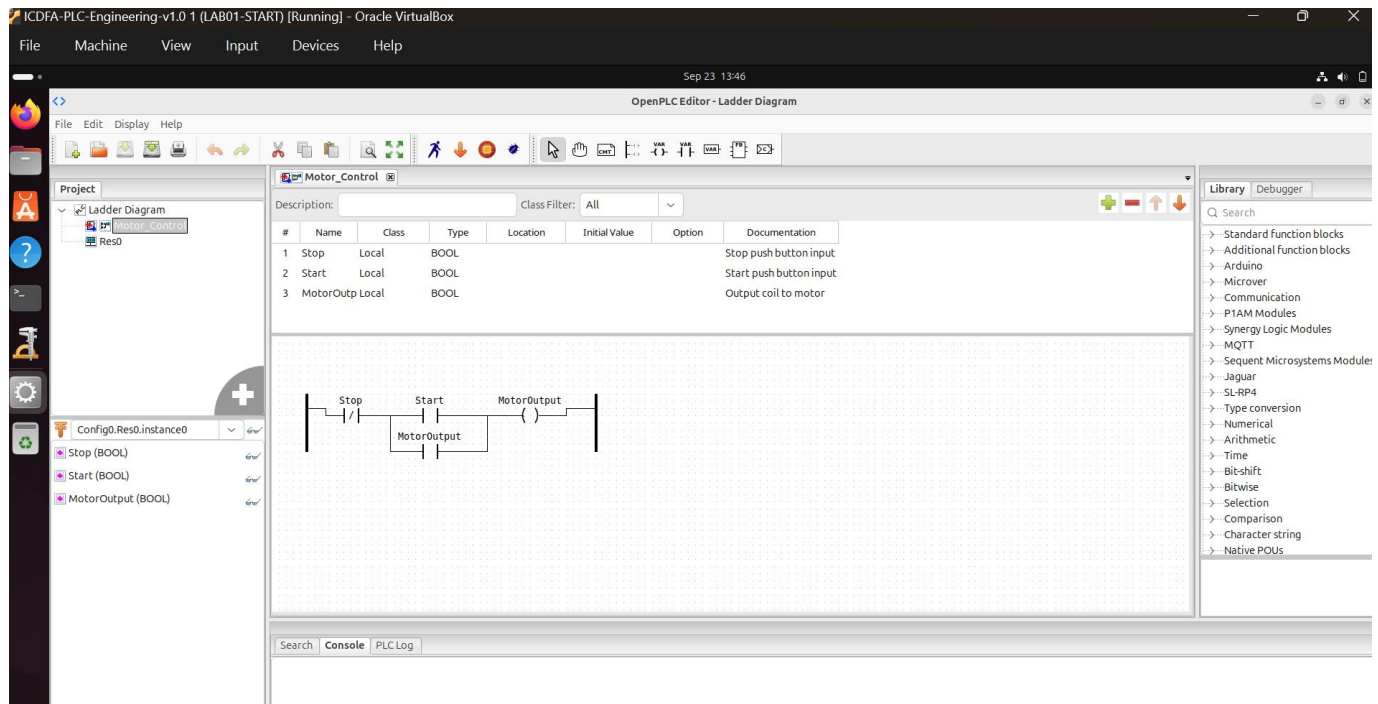
Network Interface: enp0s8

Network Type: VirtualBox Host-Only

Step 4. Open the OpenPLC Editor from the desktop/application menu.

Step 5. Select File > Open and browse to the provided PLC Programming Language Examples directory. Open the Ladder Diagram project.

Step 6. Under the Project tree, open Res0 and edit instance0 (Motor_Control). Expand the ladder diagram so the full motor-control logic is visible.



Step 7. Identify the two power rails, the rung(s), the Start contact, Stop contact, MotorOutput contact used for latching, and the MotorOutput coil.

ANSWER:

Start Contact:

The Start contact is used to initiate the motor operation. When the Start input is activated, it allows the control logic to pass through the rung and energize the MotorOutput coil, provided that the Stop condition permits operation.

Stop Contact:

The Stop contact is used to stop the motor. It is normally closed during normal operation, allowing the control circuit to remain active. When the Stop input is activated, the contact opens and interrupts the control path, causing the MotorOutput coil to de-energize.

MotorOutput Latch Contact:

The MotorOutput auxiliary contact provides the latching function. Once the MotorOutput coil is energized by pressing the Start contact, the auxiliary MotorOutput contact closes and creates an alternative path around the Start contact. This allows the motor to remain energized after the Start button is released.

MotorOutput Coil:

The MotorOutput coil represents the PLC output responsible for controlling the motor. When the conditions on the rung are satisfied, the coil becomes energized and sends the appropriate output signal to the motor starter or contactor.

Step 8. Trace the logic from the left rail to the right rail. Explain what must be true for the motor output to energize and what causes it to de-energize.

Answer:

Motor Control Logic: The motor output is energized when the Stop condition permits the circuit and the Start contact is activated. Once energized, the MotorOutput auxiliary contact provides a latching path, allowing the motor output to remain energized after the Start contact is released. The motor output is de-energized when the Stop input is activated, which breaks the control path and causes the MotorOutput coil and its latching contact to return to their normal states.

Step 9. Record one example each of a discrete input, analog input, discrete output and analog output that could appear in an industrial process.

Answer:

I/O Type	Industrial Example	What it represents
Discrete Input	Emergency-stop pushbutton	An ON/OFF signal indicating whether the emergency stop is activated.
Analog Input	Temperature sensor	A continuously varying signal representing process temperature, e.g. 4–20 mA.
Discrete Output	Motor starter/contactor	An ON/OFF command from the PLC to start or stop a motor.
Analog Output	Control valve position signal	A continuously varying signal, such as 4–20 mA, used to control valve opening.

- **A 150–250 word reflection on why network-connected PLCs create both operational benefits and cybersecurity considerations.**

ANSWER:

Network-connected Programmable Logic Controllers (PLCs) provide significant operational benefits in industrial environments because they allow automation systems to communicate with monitoring systems, engineering workstations and other operational technologies. Network connectivity can make it easier for operators and engineers to monitor processes, collect information, diagnose faults and manage equipment remotely. It can also improve coordination between different systems and support faster responses to operational problems.

However, connecting PLCs to networks also introduces cybersecurity considerations. A PLC that was previously isolated from other systems may become exposed to unauthorised access, malicious network activity or accidental changes to its configuration. A cybersecurity incident affecting a PLC could interrupt industrial operations and, depending on the process being controlled, potentially affect the availability or safe operation of equipment.

For this reason, network-connected PLC environments should be appropriately protected through measures such as access control, network segmentation, secure configuration, monitoring and regular maintenance. Users should also understand the difference between normal IT systems and industrial control environments, where availability and safe operation are particularly important. Therefore, network connectivity can improve the efficiency and visibility of industrial operations, but it should be implemented together with appropriate cybersecurity controls and operational safeguards.

Knowledge Check

1. What is the difference between an input module and an output module?

Response: An input module receives signals from field devices such as sensors, switches and pushbuttons and sends them to the PLC processor. An output module receives commands from the PLC processor and sends signals to field devices such as motors, valves, and indicators.

2. Why is a normally closed contact commonly associated with stop or emergency-stop logic?

Response: A normally closed contact allows the control circuit to remain active during normal operation. When the Stop or emergency-stop device is activated, the contact opens and breaks the control circuit, causing the motor or controlled equipment to stop.

3. Give one example of a discrete I/O signal and one example of an analog I/O signal.

Response: Discrete I/O: Emergency-stop pushbutton - provides an ON/OFF signal
Analog I/O: Temperature sensor - provides a continuously varying signal representing temperature.

4. What operational risk can arise when industrial automation devices become network-connected?

Response: Network-connected industrial automation devices can become exposed to authorised access, network attacks, or disruption of control operations. A cybersecurity incident could affect the availability or safe operation of industrial processes.